NEHA GHADGE

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Professional Summary

- ➢ Aspiring Data Scientist with hands-on experience in machine learning, deep learning, and full-stack AI deployment also in Power BI .
- ➢ Proficient in Python, SQL for database management, and data structures to optimize problem-solving and computational efficiency.
- ➢ Hands-on experience in data analysis, uncovering hidden patterns, and deriving actionable insights from datasets.
- ➢ Skilled in communicating technical insights through interactive dashboards and collaborative tools like Streamlit and LangChain.
- ➢ Continuously expanding knowledge in machine learning, deep learning, and generative AI to stay current with emerging technologies.

Technical Skills

Python	SQL	Power BI
Data structure	Machine Learning	Deep Learning
Computer Vision	Natural language processing	LLM
Data Visualization	Pandas	NumPy
Matplotlib	Scikit-learn	Generative AI
Streamlit	PyTorch / TensorFlow	Data analysis
RAG	Deepseek	Huggingface
Langchain	DAX	Agentic AI
VectorDB	Git	Dataiku

WORK EXPERIENCE

Jr. Data Scientist at IICET **(April 2025 - Present)**

➢ Contributed to a real-time project involving Exploratory Data Analysis (EDA), Data Visualization, and the development of machine learning algorithms across various domains.

➢ Gained hands-on experience with tools such as NumPy, Pandas, Matplotlib, Scikit-learn, Seaborn, TensorFlow, PyTorch, VS Code, OOP concepts, and MySQL, actively participating in the full data science lifecycle.

PROJECTS:

- **Pune House Prediction (Regression)**

Tools: Python, NumPy, Pandas, Seaborn, Matplotlib, Sci-kit learn , Linear Regressor, Streamlit , jupyter notebook
Description: In this project, I developed a predictive model to estimate house prices in Pune based on various features like location, size, number of bedrooms, amenities, and more. The goal is to provide accurate price estimates and help buyers, sellers, and investors make informed decisions.
My Role: Carried out data preprocessing, trained regression models, visualized predictions, and deployed the solution with an interactive Streamlit app.

- **Glasses and Non Glasses Prediction**

Tools: Python, TensorFlow /keras, Google Colab, OpenCV, NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, Kaggle Dataset, Data Augmentation (ImageDataGenerator)
Description: Developed a Convolutional Neural Network (CNN) model to automatically detect whether a person is wearing glasses from facial images. Trained the model on a labeled dataset of images with and without glasses, optimizing its performance for accurate classification.
My Role: Handled data preprocessing and augmentation, designed the CNN architecture, trained and tuned the model, and visualized performance metrics.

- **Cooking Recipes Chatbot using Generative AI & Vector Database (FAISS) (Generative AI)**

Tools : Python, Streamlit, FAISS, OpenAI API / Deepseek API, Langchain, Pandas, NLTK, Scikit-learn
Description: This project is an intelligent chatbot designed to assist users with cooking-related queries by providing recipe suggestions, ingredients, and step-by-step instructions. Built using Python and Streamlit for the user interface, it utilizes LangChain to orchestrate interactions between the user and the language model (OpenAI or Deepseek API). A FAISS vector database is used to store and retrieve recipe embeddings for efficient similarity search. The dataset is processed using Pandas, NLTK, and Scikit-learn for cleaning, tokenization, and embedding preparation. The chatbot ensures responses stay strictly within the cooking domain while maintaining a conversational tone.
My Role: Integrated the **LangChain framework** to manage prompt engineering and interactions with **OpenAI/Deepseek APIs**, ensuring domain-specific response generation.

PROJECTS :

MACHINE LEARNING PROJECT :

- **Network Intrusion Detection System (Classification)**

Tools: Python, NumPy, Pandas, Seaborn, Matplotlib, Sci-kit learn, Logistic Regression, Random Forest, Decision Tree, Confusion Matrix.
Description: This project aims to classify network traffic as either normal or potentially malicious using supervised machine learning techniques. The dataset used includes features extracted from network activity, allowing the model to learn patterns associated with intrusion attempts. Multiple classification algorithms such as Logistic Regression, Decision Tree, and Random Forest were trained and evaluated for performance. The final system provides a foundation for real-time intrusion detection in cybersecurity applications, helping organizations respond to threats proactively.
My Role : mplemented and trained multiple machine learning models such as Logistic Regression, Decision Tree, and Random Forest using Scikit-learn, comparing their accuracy and performance.

- **IPL Score Prediction (Regression)**

Tools: Python, NumPy, Pandas, Seaborn, Matplotlib, Sci-kit learn ,Linear Regressor, Random Forest, XGBoost, Lightgbm, Streamlit , jupyter notebook
Description: This project aims to predict the final score of an ongoing IPL (Indian Premier League) T20 cricket match using machine learning regression techniques. By analyzing various match parameters such as current score, overs bowled, wickets lost, and recent scoring trends, the model estimates the total runs a team might score by the end of their innings.
My Role: Built and evaluated multiple regression models, performed feature engineering, and deployed the solution with an interactive Streamlit interface.

- **Movie Recommendation Project (Clustering):**

Tools: Python, Pandas, Scikit-learn, Gradio, NumPy, matplotlib, seaborn, jupyter notebook .
Description: Developed a Movie Recommendation System using Cosine Similarity to recommend movies based on a given movie title. Implemented data preprocessing techniques such as one-hot encoding and normalization to enhance recommendation accuracy, with the system deployed using Gradio for a user-friendly interface.
My Role : Built and deployed a user-friendly interface using **Gradio,** enabling users to input a movie title and receive real-time recommendations.

DEEP LEARNING PROJECTS :

- **Lion and tiger Image Classification**

Tools: Python, TensorFlow /keras, Google Colab, OpenCV, NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, Kaggle Dataset, Data Augmentation (ImageDataGenerator)
Description: Developed a Lion vs. Tiger image classification model using Convolutional Neural Networks (CNN) in TensorFlow/Keras. Preprocessed and augmented image datasets, optimized model performance with techniques like dropout and batch normalization, and evaluated using accuracy, confusion matrix, and loss curves.
My Role: Evaluated model performance using accuracy score, confusion matrix, and loss/accuracy curves, and iteratively improved results through hyperparameter tuning in Google Colab.

POWER BI PROJECT:

- **HR Analytics Dashboard :**

Tools : Power BI, Power Query, DAX (Data Analysis Expressions), Interactive dashboards, charts (bar, pie, KPI cards, slicers, etc.)
Description: This project involves building an interactive HR Analytics Dashboard using Power BI to help HR departments gain insights into key workforce metrics. The dashboard visualizes data such as employee attrition, department-wise distribution, job satisfaction, performance ratings, and salary trends. It enables decision-makers to monitor workforce health, identify retention challenges, and support data-driven HR strategies. The project demonstrates how business intelligence tools can be used to transform raw HR data into actionable insights through intuitive visualizations.

My Role: Collaborated on identifying KPIs to support HR in decision-making related to employee retention, job satisfaction, and organizational performance.

- **Mobile Sales Dashboard (Using DAX) :**

  Tools : Power BI, Power Query, DAX (Data Analysis Expressions) , Interactive visualizations: Bar charts, Pie charts, KPI cards,Slicers, etc.
  Project Description: Developed an interactive HR Analytics Dashboard using Power BI to provide actionable insights into workforce dynamics. The dashboard visualizes critical HR metrics such as employee attrition, departmental distribution, job satisfaction levels, performance ratings, and salary trends. By transforming raw HR data into clear, interactive visuals, the project supports strategic decision-making, identifies retention risks, and helps monitor overall workforce health.
  My Role: Wrote DAX measures and calculated columns to compute KPIs like attrition rate, average salary, job satisfaction scores, and department-wise analysis.

Certifications & Achievements:

- Data Science and AI Specialization by IICET
- Generative AI Specialization Certificate Course.
- IBM Data Science Specialization Certified.
- Certification in C language

Education:
BCA in Computer Application
Shivaji University, Kolhapur (First Class with Distinction)